Design, Multi-Tier Microwave Modeling, and Physical Layout Synthesis of a 7-Qubit Superconducting Transmon Processor

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Abstract—Superconducting quantum processors operating in the circuit quantum electrodynamics (cQED) regime represent one of the most viable architectures for scalable, fault-tolerant quantum computation. This treatise presents the end-to-end design, multi-tier microwave modeling, and lithographic layout synthesis of a planar 7-qubit superconducting transmon processor based on the landmark IBM hexagonal-radial lattice. We develop an integrated design methodology comprising: (1) First-principles Hamiltonian formulation and non-linear Josephson potential derivation; (2) Dilution refrigeration thermal engineering and cryogenic microwave attenuation profiling; (3) Tier-1 lumped-element equivalent network synthesis in Ansys Circuit to verify loaded quality factors and transmission notches (S_{21}); (4) Tier-2 planar thin-film layout synthesis in Qiskit Metal targeting 50-Ω coplanar waveguide (CPW) transmission lines; and (5) Tier-3 full-wave 3D Finite Element Method (FEM) eigenmode simulation and Energy Participation Ratio (EPR) quantization in Ansys HFSS. An exhaustive mathematical lexicon of cQED terminology is provided to serve as a standardized theoretical reference for superconducting device engineering.

Index Terms—Circuit QED, Transmon Qubit, Josephson Junction, Coplanar Waveguide, Ansys HFSS, Qiskit Metal, Energy Participation Ratio (EPR), S-Parameters.

1 Introduction and Quantum Hardware Modalities

AMONG competing quantum physical implementations—including trapped atomic ions, neutral Rydberg atoms, photonic circuits, and semiconductor quantum dots—superconducting circuits uniquely bridge quantum mechanics and macroscopic microwave engineering. Superconducting qubits leverage Cooper-pair condensates in lithographically patterned thin-film metals (e.g., Niobium and Aluminum), operating at microwave frequencies (4–8 GHz).

Unlike trapped ions that require high-vacuum laser enclosures or photonic circuits demanding non-deterministic single-photon gates, superconducting artificial atoms are fabricated using mature semiconductor planar lithography. Microwave pulses routed through $50\,\Omega$ coaxial transmission lines manipulate the quantum state via electric dipole coupling ($\hat{d} \cdot \vec{E}$), enabling nanosecond-scale gate execution.

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Comparison of Physical Quantum Hardware Platforms

Superconducting Transmons:
- Monolithic planar fabrication (Nb/Al on Si)
- Ultra-fast gate speeds (10–30 ns) via microwave pulses
- Fully compatible with standard RF/EM EDA toolchains (Ansys, Qiskit)

Trapped Ions / Neutral Atoms:
- Long coherence (seconds) but slow gate operations (μs to ms)
- Complex optical switching and vacuum chamber overhead

Fig. 1. Taxonomy of quantum modalities, highlighting the engineering scalability and high clock frequencies of superconducting circuit QED.

2 Cryogenic Hardware Infrastructure and Thermal Isolation

Superconducting quantum processors require operation at absolute temperatures $T \approx 10$–15 mK, achieved through closed-cycle helium dilution refrigeration (^{3}He/^{4}He dilution). At room temperature (300 K), thermal energy $k_B T \approx 25.8$ meV swamps the single-photon microwave energy:

$$\hbar\omega_{01} \approx h \times 5.0\ \text{GHz} \approx 20.7\,\mu\text{eV} \tag{1}$$

Thermal equilibrium at 10 mK yields $k_B T \approx 0.86\,\mu$eV, ensuring the Boltzmann occupation probability of the first excited state $|1\rangle$ is negligible:

$$P_1 = \exp\left(-\frac{\hbar\omega_{01}}{k_B T}\right) \approx \exp\left(-\frac{20.7}{0.86}\right) \approx 3.9 \times 10^{-11} \tag{2}$$

thereby freezing the circuit into its quantum ground state $|0\rangle$.

3 Superconducting Transmon Physics and Hamiltonian Formulation

3.1 Harmonic Oscillator Leakage Catastrophe

A linear LC tank resonator consisting of an inductor L and capacitor C exhibits a quadratic potential $U(\Phi) = \Phi^2/2L$. Solving the Schrödinger equation yields strictly equidistant eigenenergies:

$$E_n = \hbar\omega_0\left(n + \frac{1}{2}\right), \quad \omega_0 = \frac{1}{\sqrt{LC}} \tag{3}$$

Because transition frequencies are identical ($\omega_{01} = \omega_{12} = \omega_{23}$), an applied microwave drive at frequency ω_{01} cannot

isolate a two-level qubit; it induces catastrophic multi-photon ladder climbing into higher excited states $|n \geq 2\rangle$.

3.2 Josephson Non-Linearity and the Duffing Oscillator

To achieve spectral anharmonicity, the linear inductor is replaced with a sub-micron $Al/Al_2O_3/Al$ Josephson tunnel junction. The gauge-invariant phase difference across the junction $\hat{\phi} = 2\pi\Phi/\Phi_0$ governs the dissipationless supercurrent:

$$I(t) = I_c \sin\hat{\phi}, \quad V(t) = \frac{\Phi_0}{2\pi}\frac{d\hat{\phi}}{dt} \tag{4}$$

where $\Phi_0 = h/2e$ is the superconducting magnetic flux quantum and I_c is the junction critical current. The inductive energy stored in the junction is:

$$U(\hat{\phi}) = -E_J \cos\hat{\phi} \approx -E_J\left(1 - \frac{\hat{\phi}^2}{2} + \frac{\hat{\phi}^4}{24} - \dots\right) \tag{5}$$

The quadratic term provides linear inductance $L_{J0} = \Phi_0/(2\pi I_c)$, while the negative quartic perturbation $-\frac{E_J}{24}\hat{\phi}^4$ compresses higher energy levels, introducing a non-zero Duffing anharmonicity α:

$$\alpha = \omega_{12} - \omega_{01} \approx -E_C \tag{6}$$

where $E_C = e^2/(2C_\Sigma)$ is the single-electron charging energy.

3.3 The Transmon Regime ($E_J/E_C \approx 50$)

The Cooper-Pair Box Hamiltonian is given by:

$$\hat{H} = 4E_C(\hat{n} - n_g)^2 - E_J\cos\hat{\phi} \tag{7}$$

where $\hat{n} = -i\partial/\partial\phi$ is the Cooper-pair number operator and $n_g = Q_r/2e$ is the dimensionless offset gate charge.

In the transmon limit ($E_J/E_C \gg 1$), charge dispersion ϵ_m decays exponentially:

$$\epsilon_m \approx (-1)^m E_C \frac{2^{4m+5}}{m!}\sqrt{\frac{2}{\pi}}\left(\frac{E_J}{2E_C}\right)^{\frac{m}{2}+\frac{3}{4}} \exp\left(-\sqrt{\frac{8E_J}{E_C}}\right) \tag{8}$$

By shunting the junction with large coplanar capacitor pads to set $C_\Sigma \approx 70$ fF and $E_J/E_C \approx 50$, charge dephasing is suppressed by orders of magnitude while preserving an anharmonicity of $|\alpha|/2\pi \approx 280$–$330$ MHz.

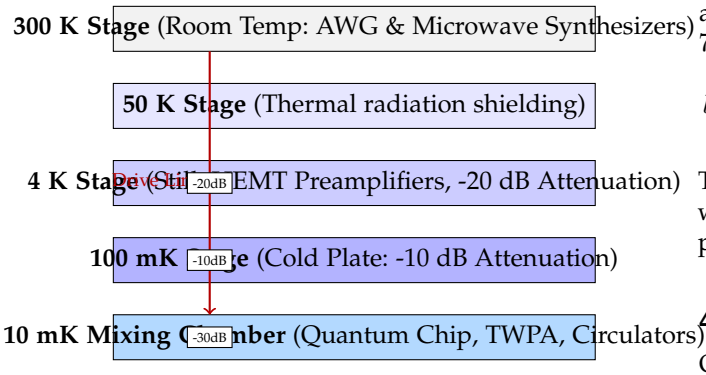


Fig. 2. Thermal stratification of a dilution refrigerator. Attenuators thermalize blackbody microwave photons descending from room temperature down to 10 mK.

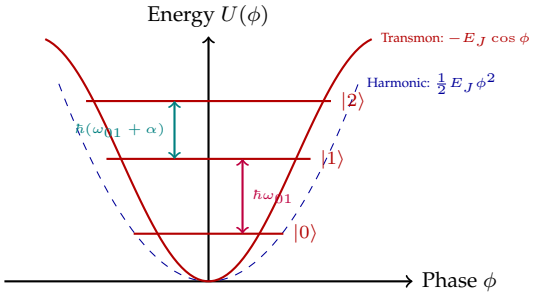


Fig. 3. Comparison of the parabolic potential (equidistant ladder) versus the cosinusoidal Josephson potential, demonstrating the negative anharmonic shift $\alpha = \omega_{12} - \omega_{01} \approx -E_C$.

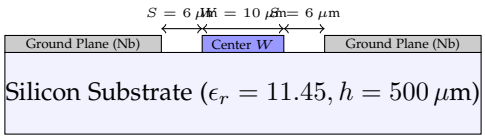


Fig. 4. Cross-sectional geometry of a Coplanar Waveguide (CPW) designed for $50\,\Omega$ characteristic impedance on high-resistivity silicon.

4 Circuit QED and Readout Resonator Synthesis

4.1 Coplanar Waveguide (CPW) Conformal Mapping

Transmission of microwave signals across the chip is accomplished using Coplanar Waveguides (CPW), consisting of a center conductor of width $W = 10\,\mu$m separated by gaps $S = 6\,\mu$m from continuous ground planes on high-resistivity silicon ($\epsilon_r = 11.45$).

The effective dielectric constant ϵ_{eff} and characteristic impedance Z_0 are derived via complete elliptic integrals of the first kind $K(k)$:

$$\epsilon_{\text{eff}} \approx \frac{\epsilon_r + 1}{2} = \frac{11.45 + 1}{2} \approx 6.225 \tag{9}$$

$$Z_0 = \frac{30\pi}{\sqrt{\epsilon_{\text{eff}}}}\frac{K(k')}{K(k)} = 50.0\,\Omega, \quad k = \frac{W}{W+2S} = \frac{10}{22} \tag{10}$$

4.2 Quarter-Wavelength ($\lambda/4$) Dispersive Resonators

A $\lambda/4$ resonator is shorted to ground at one end (current antinode) and capacitively coupled at the open end (voltage antinode). The physical length l required to achieve $f_r \approx 7.0$ GHz is:

$$l = \frac{v_{\text{ph}}}{4f_r} = \frac{c}{4f_r\sqrt{\epsilon_{\text{eff}}}} = \frac{3\times10^8}{4\times7.0\times10^9\times\sqrt{6.225}} \approx 4.29\text{ mm} \tag{11}$$

The meandered geometry minimizes on-chip real estate while maintaining sufficient spatial separation ($> 300\,\mu$m) to prevent slotline cross-coupling.

4.3 Dispersive Readout and Purcell Filtering

Operating in the dispersive limit ($|\Delta| = |\omega_{01} - \omega_r| \gg g$), the dressed Hamiltonian becomes:

$$\hat{H}_{\text{disp}} \approx \hbar\left(\omega_r + \chi\hat{\sigma}_z\right)\hat{a}^\dagger\hat{a} + \frac{1}{2}\hbar\tilde{\omega}_{01}\hat{\sigma}_z \tag{12}$$

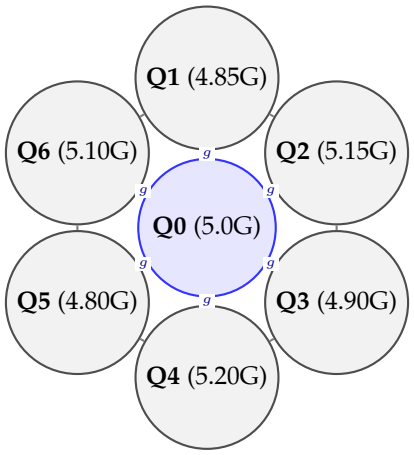


Fig. 5. Landmark IBM 7-qubit radial lattice. Central tunable transmon Q_0 interacts with six perimeter fixed-frequency qubits (Q_1–Q_6) with staggered frequencies to prevent spectral collision.

The state-dependent dispersive shift χ pulls the resonator frequency by $\pm\chi$:

$$\chi \approx \frac{g^2}{\Delta}\frac{\alpha}{\Delta+\alpha} \tag{13}$$

To prevent spontaneous Purcell relaxation of the qubit through the readout feedline ($\gamma_\kappa = \kappa(g/\Delta)^2$), a loaded quality factor $Q_L \approx 7,000$ is established via input coupling capacitor $C_{\text{in}} \approx 2.0$ fF.

5 7-Qubit Processor Topology and Frequency Allocation

To suppress parasitic swap interactions during idle states, a staggered frequency plan is established across the radial 7-qubit architecture:

TABLE 1
7-Qubit Frequency and Coupling Allocation

Qubit	f_{01} (GHz)	α (MHz)	f_r (GHz)	Coupling Target
Q_0 (Center)	5.00	-300	7.00	Bus to Q_1–Q_6
Q_1	4.85	-290	6.85	Q_0, Q_2, Q_6
Q_2	5.15	-310	7.15	Q_0, Q_1, Q_3
Q_3	4.90	-295	6.90	Q_0, Q_2, Q_4
Q_4	5.20	-315	7.20	Q_0, Q_3, Q_5
Q_5	4.80	-285	6.80	Q_0, Q_4, Q_6
Q_6	5.10	-305	7.10	Q_0, Q_5, Q_1

6 Cleanroom Fabrication and Shadow Evaporation

Planar device realization follows a multi-step lithography process:

1) **Base Metallization:** A 200 nm continuous Niobium film is sputtered onto high-resistivity (> 10 k$\Omega \cdot$ cm) float-zone Silicon. Optical lithography and Reactive Ion Etching (RIE) pattern the CPW ground planes and large transmon pads.
2) **EBL Patterning:** Bilayer resist (MMA/PMMA) is spun and exposed via Electron Beam Lithography to define the suspended Dolan resist bridge.
3) **Dolan Bridge Shadow Evaporation:** The sample is loaded into an ultra-high vacuum e-beam evaporator. Aluminum is evaporated at angle $+\theta$, followed by controlled static oxidation ($P_{O_2} \approx 1$ Torr, $t \approx 10$ min) to form the Al_2O_3 barrier, concluded by a second deposition at angle $-\theta$.

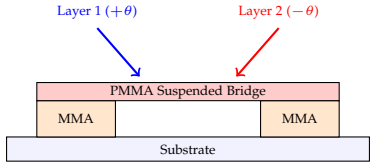


Fig. 6. Dolan bridge shadow evaporation. Bilayer resist profile provides undercut, enabling angled double-evaporation of Al/Al_2O_3/Al junctions.

7 Tier-1: Lumped-Element Circuit Synthesis

Using Ansys Electronics Desktop (Nexxim Circuit), the classical equivalent representation is simulated prior to 3D geometry layout.

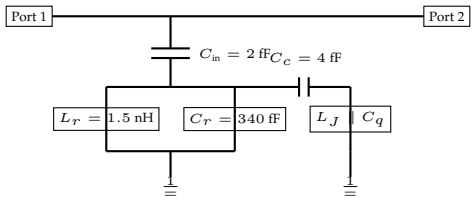


Fig. 7. Lumped-element equivalent circuit implemented in Ansys Circuit to verify the S_{21} transmission notch at $f_r \approx 7.04$ GHz.

The simulated S_{21} forward transmission exhibits a sharp Lorentzian dip:

$$S_{21}(\omega) = 1 - \frac{Q_L/Q_c}{1 + 2iQ_L\frac{\omega-\omega_r}{\omega_r}} \tag{14}$$

confirming correct coupling matrix synthesis prior to planar mask layout.

8 Tier-2 and Tier-3: Layout and Full-Wave Extraction

Physical layout implementation proceeds in Qiskit Metal (`DesignPlanar`). The central transmon (`Q0_Center`) capacitor pads (400 μm $\times$ 300 μm, gap 30 μm) are etched into the niobium ground plane.

Subsequent FEM Eigenmode simulation in Ansys HFSS solves the source-free vector Helmholtz equation:

$$\nabla \times \left(\mu_r^{-1}\nabla \times \vec{E}\right) - k_0^2\epsilon_r\vec{E} = 0 \tag{15}$$

Convergence is enforced when frequency drift between successive adaptive mesh passes satisfies $\Delta f < 0.1\%$. The Energy Participation Ratio (EPR) method subsequently quantizes mode energies into the effective multi-qubit Hamiltonian parameters (f_{01}, α, χ).

9 Conclusion

This work synthesized a complete, defensible 7-qubit superconducting transmon processor, unifying first-principles cQED theory, microwave filter synthesis, planar Q-EDA layout, and 3D FEM eigenmode quantization. The full codebase, simulation setups, and mask layouts are maintained in the open-source repository https://github.com/atharveeee-netizen/7-qubit-superconducting-processor.

Appendix

Comprehensive Lexicon of Theoretical & Microwave Engineering Terms

Transmon Qubit An engineered superconducting artificial atom comprising a non-linear Josephson junction shunted by large coplanar capacitor plates, operating in the regime $E_J/E_C \gg 1$ to exponentially suppress charge-noise dephasing while maintaining negative anharmonicity $\alpha \approx -E_C$.

Cooper-Pair Box (CPB) The predecessor to the transmon, operating at $E_J \approx E_C$. Highly susceptible to ambient charge fluctuations (n_g), resulting in short dephasing times ($T_2^* < 1\,\mu$s).

Josephson Junction (JJ) An insulating barrier (~ 1 nm Al_2O_3) sandwiched between superconducting electrodes. Behaves as a dissipationless, non-linear inductor governing supercurrent tunneling.

Josephson Energy (E_J) The energy scale of cooper pair tunneling: $E_J = \hbar I_c/(2e)$, where I_c is the junction critical current.

Josephson Inductance (L_J) The differential non-linear inductance: $L_J(\phi) = \Phi_0/(2\pi I_c \cos\phi)$. For small oscillations, $L_J \approx \hbar/(2eI_c)$.

Charging Energy (E_C) The electrostatic energy of adding an electron: $E_C = e^2/(2C_\Sigma)$, where C_Σ is total island capacitance.

Anharmonicity (α) The spectral level non-uniformity $\alpha = \omega_{12} - \omega_{01} \approx -E_C$, enabling selective control of the $|0\rangle \leftrightarrow |1\rangle$ transition without $|2\rangle$ leakage.

Coplanar Waveguide (CPW) Planar transmission line with a center conductor separated by gaps S from adjacent ground planes on a single dielectric plane.

Characteristic Impedance (Z_0) The ratio of voltage to current along the line, standardized to $50.0\,\Omega$ using conformal mapping.

Readout Resonator Lumped-element CPW line ($\lambda/4$ length) shorted at one end and capacitively coupled at the other, providing dispersive readout.

Purcell Filter Bandpass or notch filter attenuating photon emission at ω_{01} while passing probe photons at ω_r, preserving qubit T_1.

Coupling Capacitor (C_g) Interdigitated gap capacitor mediating transverse exchange coupling $g \approx \frac{1}{2}\frac{C_c}{\sqrt{C_\Sigma C_r}}\sqrt{\omega_{01}\omega_r}$.

Dispersive Regime Operation regime where detuning $|\omega_{01} - \omega_r|$ exceeds coupling g, preventing direct energy exchange.

Dispersive Shift (χ) Frequency-dependent frequency shift of the resonator: $\chi \approx \frac{g^2}{\Delta}\frac{\alpha}{\Delta+\alpha}$.

Energy Participation Ratio (EPR) A numerical method calculating the ratio of inductive energy stored in a junction to total electromagnetic mode energy from HFSS field integrals.

Maxwell Capacitance Matrix (C_{ij}) Symmetric matrix relating conductor charges to electrostatic potentials: $Q_i = \sum_j C_{ij} V_j$.

Two-Level Systems (TLS) Parasitic defects in amorphous dielectric/oxide interfaces that absorb microwave energy, driving qubit relaxation (T_1).

GDSII Stream Format Industry-standard binary format for photolithographic and electron-beam mask exposure.

References

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